

Time Series Econometrics

Persistence, Trends, Forecasting, and Dynamic Systems

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Roadmap

1. Motivation and Language
2. Stationarity and Dependence
3. AR, MA, and ARMA Models
4. Trends and Unit Roots
5. Forecasting
6. Time-Series Regression and Distributed Lags
7. VARs and Dynamic Systems
8. Applications and Takeaways

Why time series econometrics is different

Cross-sectional rows are often exchangeable; dated observations are not

The order of quarterly inflation, monthly unemployment, daily exchange rates, or annual GDP is part of the economic information.

Questions about dynamics

- Is today's inflation related to yesterday's inflation?
- Do output shocks fade or permanently shift the level?
- Do interest rates help forecast future inflation?

New risks

- serial dependence
- stochastic and deterministic trends
- regime changes and breaks
- spurious levels regressions

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Central question

When data are ordered in time, how do dependence, persistence, trends, and dynamic feedback change the way we model economic variables?

Three data structures, three comparison problems

Data	Typical observation	Main comparison	Dependence issue
Cross-section	household, worker, firm, country at one date	across units	omitted heterogeneity, clustering
Time series	one aggregate or market over many dates	across dates	persistence, trends, breaks
Panel	many units over many dates	across and within units	unit heterogeneity plus serial dependence

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Time-series perspective

Dependence over time is not merely a nuisance. Inflation persistence, exchange-rate predictability, policy transmission, and business-cycle propagation are all questions about how information moves through time.

Forecasting, description, and causality

Forecasting

Use information at date T
to predict

$$Y_{T+h}.$$

Dynamic description

Summarize persistence,
adjustment, and lead-lag
patterns.

Causal analysis

Estimate how an
economically meaningful
shock changes future
outcomes.

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Summarize persistence, adjustment, and lead-lag patterns.

Causal analysis

Estimate how an economically meaningful shock changes future outcomes.

Do not confuse the three

A variable may help forecast another variable without representing an exogenous causal shock. A VAR can summarize dynamic correlations without identifying monetary-policy transmission.

Stochastic process versus realization

Definition

A **stochastic process** is a sequence of random variables

$$\{Y_t : t \in \mathcal{T}\}$$

defined on a common probability space. The observed time series

$$\{Y_t : t = 1, \dots, T\}$$

is one realized path from that process.

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What we observe

One path for U.S. inflation, euro-area unemployment, an exchange rate, or an interest rate.

What the model asks

What probabilistic mechanism could have produced this path, and what does the path teach us about persistence or prediction?

Information sets and innovations

Time-series conditioning

Let

$$\mathcal{F}_t = \sigma(Y_t, Y_{t-1}, Y_{t-2}, \dots)$$

denote the information generated by the history of Y . A forecast made at time T is naturally written as

$$\hat{Y}_{T+h|T} = \mathbb{E}(Y_{T+h} \mid \mathcal{F}_T).$$

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Predictable component plus innovation

$$Y_t = \mathbb{E}(Y_t \mid \mathcal{F}_{t-1}) + \varepsilon_t, \quad \mathbb{E}(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0.$$

The innovation is the part of Y_t not predictable from the past under the maintained model.

Lag notation records economic timing

Lag operator

$$LY_t = Y_{t-1}, \quad L^h Y_t = Y_{t-h}.$$

The first difference is

$$\Delta Y_t = Y_t - Y_{t-1} = (1 - L)Y_t.$$

Growth rates

If $Y_t > 0$,

$$\Delta \log Y_t = \log Y_t - \log Y_{t-1}$$

is approximately the growth rate.

Timing is substantive

Whether a regressor is contemporaneous, lagged, or forward-looking changes the interpretation and the exogeneity assumptions.

Dependence over time: autocovariance and ACF

Autocovariance

For a process with finite second moments,

$$\gamma_t(h) = \text{Cov}(Y_t, Y_{t-h}).$$

If this covariance depends only on the lag h , not on calendar time t , write it as γ_h .

Dependence over time: autocovariance and ACF

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Autocorrelation function

$$\rho_h = \frac{\gamma_h}{\gamma_0}, \quad \gamma_0 = \text{Var}(Y_t).$$

The ACF is unit-free, $\rho_0 = 1$, and $-1 \leq \rho_h \leq 1$.

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Interpretation

Autocorrelation is a summary of linear persistence. It is the first diagnostic language for time-series dynamics.

Strict and weak stationarity

Strict stationarity

For every set of dates $t_1, \dots, t_m$ and every shift h , the joint distribution of

$$(Y_{t_1}, \dots, Y_{t_m})$$

is the same as that of

$$(Y_{t_1+h}, \dots, Y_{t_m+h}).$$

Weak stationarity

$$\mathbb{E}(Y_t) = \mu,$$

$$\text{Var}(Y_t) = \gamma_0 < \infty,$$

$$\text{Cov}(Y_t, Y_{t-h}) = \gamma_h.$$

Only the first two moments and autocovariances must be stable.

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Main applied meaning

Stationarity is a stability restriction. It does not say the realized series is flat; it says the probabilistic environment is stable over time.

White noise and shocks

White noise

A process ε_t is white noise if

$$\mathbb{E}(\varepsilon_t) = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2, \quad \text{Cov}(\varepsilon_t, \varepsilon_s) = 0 \quad (t \neq s).$$

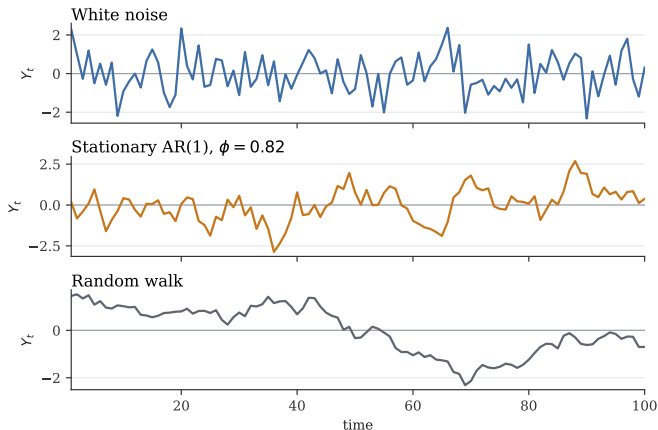
Why it matters

White-noise shocks are the unpredictable input into ARMA, regression, and VAR models.

What it does not imply

Zero autocorrelation is not the same as full independence. A series can have uncorrelated returns but persistent volatility.

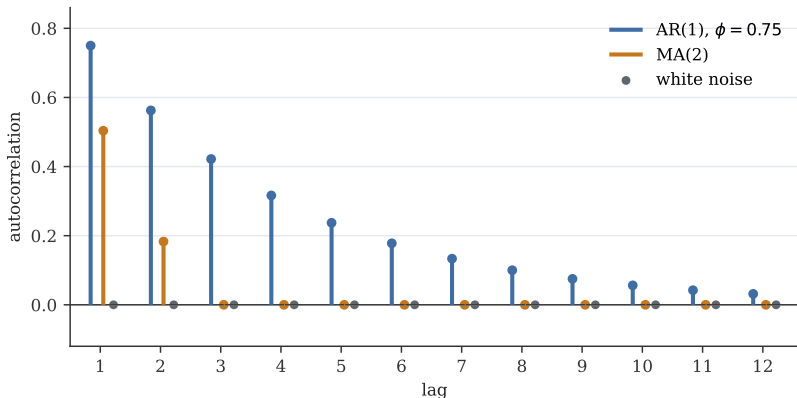
Three schematic paths



Visual distinction

White noise has no linear memory. A stationary persistent process moves around a stable distribution. A random walk carries accumulated shocks in its level.

ACF patterns are a compact diagnostic



Reading the plot

Fast decay suggests weak persistence. Slow decay suggests high persistence or possible nonstationarity. Alternating signs suggest negative dynamic feedback.

The AR(1) model

Model

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

Mean under stationarity

If $\mathbb{E}(Y_t) = \mathbb{E}(Y_{t-1}) = \mu$,

$$\mu = c + \phi\mu, \quad \mu = \frac{c}{1 - \phi}.$$

Stability

For AR(1), a finite stationary distribution requires

$$|\phi| < 1.$$

The parameter ϕ measures persistence.

AR(1) Algebra

Centering reveals shock propagation

With $\mu = c/(1 - \phi)$, define $y_t = Y_t - \mu$. Then

$$y_t = \phi y_{t-1} + \varepsilon_t.$$

Repeated substitution gives, when $|\phi| < 1$,

$$y_t = \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \cdots .$$

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A one-unit innovation today

effect on $y_t = 1$, effect on $y_{t+1} = \phi$, effect on $y_{t+h} = \phi^h$.

Persistence

If $\phi = 0.2$, the shock disappears quickly. If $\phi = 0.98$, the shock remains important many periods later.

AR(1) variance and autocorrelation

Variance

For the stationary centered process,

$$\gamma_0 = \text{Var}(y_t) = \phi^2 \gamma_0 + \sigma^2,$$

so

$$\text{Var}(Y_t) = \frac{\sigma^2}{1 - \phi^2}.$$

ACF

For $h \geq 1$,

$$\gamma_h = \phi \gamma_{h-1},$$

so

$$\rho_h = \phi^h.$$

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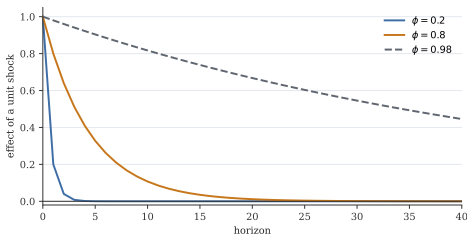
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Interpreting the AR(1) coefficient

Value of ϕ	Interpretation
$\phi = 0$	no persistence beyond the current shock
$0 < \phi < 1$	positive persistence; shocks decay monotonically
ϕ close to 1	high persistence; shocks decay slowly
$-1 < \phi < 0$	alternating signs after a shock
$\phi = 1$	unit root; shocks have permanent effects on the level
$ \phi > 1$	explosive behavior in the linear model

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Half-life

The half-life of a shock is the h such that $|\phi|^h = 1/2$:

$$h = \frac{\log(1/2)}{\log |\phi|}.$$

If $\phi = 0.9$, the half-life is about 6.6 periods; if $\phi = 0.98$, it is about 34.3 periods.

Inflation persistence

A starting model

$$\pi_t = c + \phi_1 \pi_{t-1} + \dots + \phi_p \pi_{t-p} + \varepsilon_t.$$

Economic question

Do inflation shocks die out quickly, or does inflation remain elevated after a disturbance?

Why it matters

Persistence shapes central-bank credibility, sacrifice ratios, and the expected cost of disinflation.

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Empirical caution

Inflation regimes change. A sample spanning high inflation, inflation targeting, supply shocks, and policy-framework changes may not be described by one stable AR model.

AR(p) models and lag polynomials

Autoregression of order p

$$Y_t = c + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \varepsilon_t.$$

In lag notation,

$$\Phi(L)Y_t = c + \varepsilon_t, \quad \Phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p.$$

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Why more than one lag?

Multiple lags capture momentum, delayed adjustment, reversal, and policy smoothing.

Stationarity intuition

The roots of

$$\Phi(z) = 0$$

must lie outside the unit circle. No lag dynamic should create a unit root or explosive root.

Root Condition

MA(q): finite shock memory

Moving-average model

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

Equivalently,

$$Y_t - \mu = \Theta(L)\varepsilon_t, \quad \Theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q.$$

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MA(1)

If $Y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}$, then

$$\text{Var}(Y_t) = \sigma^2(1 + \theta^2), \quad \rho_1 = \frac{\theta}{1 + \theta^2}, \quad \rho_h = 0 \quad (h \geq 2).$$

ARMA models combine persistence and shock structure

ARMA(p, q)

$$\Phi(L)(Y_t - \mu) = \Theta(L)\varepsilon_t,$$

where

$$\Phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p, \quad \Theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q.$$

Model	Equation	Main idea
AR(p)	$\Phi(L)(Y_t - \mu) = \varepsilon_t$	persistence through lagged values
MA(q)	$Y_t - \mu = \Theta(L)\varepsilon_t$	finite propagation of shocks
ARMA(p, q)	$\Phi(L)(Y_t - \mu) = \Theta(L)\varepsilon_t$	persistence plus serially structured innovations

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Use

ARMA models are natural for stationary series when the goal is compact description or forecasting.

Deterministic trends

Trend-stationary process

$$Y_t = \alpha + \delta t + u_t,$$

where u_t is stationary.

Interpretation

The mean changes deterministically with t , but deviations from the trend are stationary.

Shock behavior

A shock moves the series away from trend temporarily if u_t is stationary. The long-run path is still anchored by $\alpha + \delta t$.

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Transformation

The natural operation is detrending: estimate or model the deterministic trend and study the stationary residual component.

Random walks and unit roots

Random walk

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Iterating backward,

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No fixed variance

If $\text{Var}(\varepsilon_t) = \sigma^2$, then

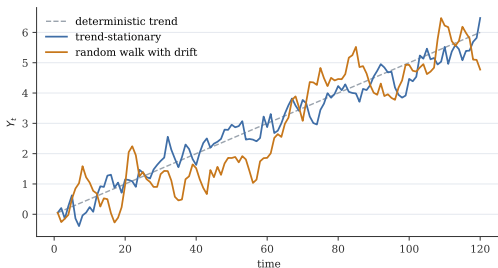
$$\text{Var}(Y_t \mid Y_0) = t\sigma^2.$$

The level is not weakly stationary.

Permanent shocks

A shock at date t enters every future level through the accumulated sum.

Trend-stationary versus difference-stationary



Feature	Trend-stationary	Difference-stationary
Canonical model	$Y_t = \alpha + \delta t + u_t$	$Y_t = Y_{t-1} + c + \varepsilon_t$
Stationary object	deviation from trend	first difference ΔY_t
Shock effect	temporary	permanent
Transformation	detrend	difference

Integrated processes and transformations

Difference stationarity

If Y_t follows a random walk with drift,

$$Y_t = c + Y_{t-1} + \varepsilon_t,$$

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$$\Delta Y_t = c + \varepsilon_t.$$

The differenced series may be stationary even though the level is not.

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$I(1)$ and $I(0)$

Y_t is integrated of order one, written $Y_t \sim I(1)$, if Y_t is nonstationary but ΔY_t is stationary. A stationary process is $I(0)$.

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Do not difference mechanically

Differencing changes the question. If Y_t is log GDP, then $\Delta \log Y_t$ is growth. A model for GDP growth is not a model for the level of GDP.

Dickey–Fuller intuition

Start with the AR(1)

$$Y_t = \rho Y_{t-1} + \varepsilon_t.$$

Subtract Y_{t-1} :

$$\Delta Y_t = (\rho - 1)Y_{t-1} + \varepsilon_t = \pi Y_{t-1} + \varepsilon_t.$$

Null

$$H_0 : \rho = 1 \iff H_0 : \pi = 0.$$

There is a unit root.

Stationary alternative

$$H_1 : |\rho| < 1,$$

often written as

$$H_1 : \pi < 0.$$

ADF Details

ADF regressions: deterministic terms and short-run dynamics

Augmented Dickey–Fuller regression

$$\Delta Y_t = \alpha + \delta t + \pi Y_{t-1} + \sum_{j=1}^k a_j \Delta Y_{t-j} + e_t.$$

Why add lagged differences?

They absorb short-run serial correlation so that e_t is closer to white noise.

Why special critical values?

Under the unit-root null, Y_{t-1} is nonstationary, so the usual t distribution does not apply.

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Applied judgment

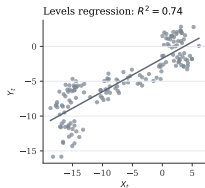
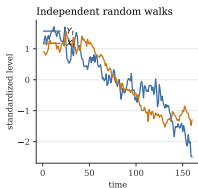
Failing to reject a unit root does not prove a unit root. Tests can have low power against highly persistent stationary alternatives.

Spurious regression

Two unrelated random walks

$$Y_t = Y_{t-1} + \varepsilon_t, \quad X_t = X_{t-1} + \nu_t,$$

with independent shocks.



The trap

The levels regression

$$Y_t = \alpha + \beta X_t + u_t$$

can show a high R^2 and a significant $\hat{\beta}$ even though the series are unrelated.

Cointegration preview

The important exception

If Y_t and X_t are each $I(1)$ but a linear combination is stationary,

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Economic meaning

The levels share a stable long-run relation even though each level is individually nonstationary. Examples might include two interest rates or related prices whose spread is mean-reverting.

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Error-correction intuition

$$\Delta Y_t = \alpha + \lambda(Y_{t-1} - \beta X_{t-1}) + \Gamma(L)\Delta X_t + e_t.$$

If $\lambda < 0$, a positive deviation from the long-run relation predicts downward adjustment.

Forecasts as conditional expectations

Forecast and forecast error

$$\hat{Y}_{T+h|T} = \mathbb{E}(Y_{T+h} \mid \mathcal{F}_T), \quad e_{T+h} = Y_{T+h} - \hat{Y}_{T+h|T}.$$

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Under squared-error loss

The conditional expectation is the optimal forecast. The forecast is always conditional on an information set and a maintained model.

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Interpretation

Forecasting is not guessing a fixed future. It is summarizing the distribution of future outcomes given what is known at the forecast origin.

AR(1) forecasts

For

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad |\phi| < 1,$$

the one-step forecast is

$$\hat{Y}_{T+1|T} = c + \phi Y_T.$$

Using $\mu = c/(1 - \phi)$,

$$\hat{Y}_{T+h|T} = \mu + \phi^h (Y_T - \mu).$$

AR(1) forecasts

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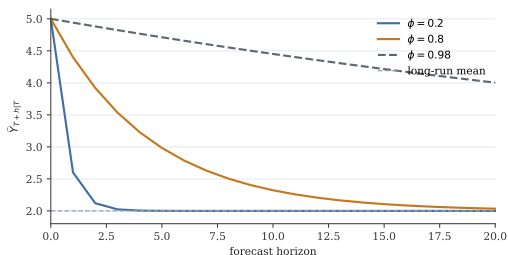
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Persistence controls forecast reversion

When $|\phi| < 1$, forecasts converge to μ as the horizon grows. Higher ϕ means slower convergence.

Forecast uncertainty

h -step AR(1) forecast error

$$Y_{T+h} - \hat{Y}_{T+h|T} = \varepsilon_{T+h} + \phi\varepsilon_{T+h-1} + \cdots + \phi^{h-1}\varepsilon_{T+1}.$$

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Forecast error variance

$$\text{Var}(e_{T+h}) = \sigma^2 \sum_{j=0}^{h-1} \phi^{2j} = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2}.$$

Forecast uncertainty

h -step AR(1) forecast error

$$Y_{T+h} - \hat{Y}_{T+h|T} = \varepsilon_{T+h} + \phi\varepsilon_{T+h-1} + \cdots + \phi^{h-1}\varepsilon_{T+1}.$$

Forecast error variance

$$\text{Var}(e_{T+h}) = \sigma^2 \sum_{j=0}^{h-1} \phi^{2j} = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2}.$$

Lesson

Forecast uncertainty rises with the horizon and approaches the unconditional variance when $|\phi| < 1$.

Forecasting practice

Modeling choices

- levels, logs, growth rates, or differences
- lag length
- deterministic terms and seasonality
- parameter stability

Evaluation choices

- out-of-sample performance
- benchmark model
- point forecast versus uncertainty
- purpose: prediction, stress testing, or policy

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Discipline

A forecast is conditional on a model and an information set. Good applied work states both.

Regression with time-series data

Static regression

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This often says too much: the effect of X_t is contemporaneous and complete within the same period.

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Why dynamics matter

Policy rates affect borrowing costs, spending, unemployment, and inflation over several quarters.

Exogeneity is about timing

If today's shock affects tomorrow's regressor, strict exogeneity fails even if the regressor is predetermined in some weaker sense.

Serial correlation in regression errors

Example

$$u_t = \rho u_{t-1} + e_t.$$

Why it matters

1. It can signal omitted dynamics.
2. It affects standard errors and inference.
3. With lagged dependent variables, it can threaten consistency.

Applied response

Use dynamic specifications, residual diagnostics, HAC standard errors when appropriate, or explicit error models.

Distributed lag models

Finite distributed lag

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \cdots + \beta_q X_{t-q} + u_t.$$

Dynamic effects

- β_0 : impact effect
- β_j : effect associated with the j th lag
- $\sum_{j=0}^q \beta_j$: long-run multiplier for a permanent change in X

Example

With L_t the lending rate and i_t the policy rate,

$$L_t = \alpha + \beta_0 i_t + \beta_1 i_{t-1} + \beta_2 i_{t-2} + u_t.$$

Pass-through may be gradual.

ADL models and long-run multipliers

ADL(1,1)

$$Y_t = \alpha + \rho Y_{t-1} + \beta_0 X_t + \beta_1 X_{t-1} + u_t.$$

The lagged dependent variable captures persistence in Y .

ADL models and long-run multipliers

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$$Y_t = \alpha + \rho Y_{t-1} + \beta_0 X_t + \beta_1 X_{t-1} + u_t.$$

The lagged dependent variable captures persistence in Y .

Long-run multiplier

For a permanent change in X , set $Y_t = Y_{t-1} = Y$ and $X_t = X_{t-1} = X$:

$$Y = \alpha + \rho Y + (\beta_0 + \beta_1)X.$$

Thus

$$\frac{\partial Y}{\partial X} = \frac{\beta_0 + \beta_1}{1 - \rho}, \quad |\rho| < 1.$$

LRM Derivation

Dynamic regression workflow

1. Plot the series and decide on levels, logs, differences, or growth rates.
2. Check persistence, seasonality, and obvious breaks.
3. Start with an economically motivated lag structure.
4. Inspect residual autocorrelation.
5. Interpret impact, delayed, and long-run effects separately.
6. State whether the exercise is predictive, descriptive, or causal.

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Common mistake

A lag coefficient is not automatically causal. It is a dynamic conditional association unless timing, omitted variables, policy rules, and shock structure justify a causal interpretation.

A two-variable VAR

Let

$$Z_t = \begin{pmatrix} y_t \\ x_t \end{pmatrix}.$$

A VAR(1) is

$$Z_t = a + A_1 Z_{t-1} + e_t,$$

or

$$y_t = a_y + a_{11}y_{t-1} + a_{12}x_{t-1} + e_{yt},$$

$$x_t = a_x + a_{21}y_{t-1} + a_{22}x_{t-1} + e_{xt}.$$

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Why economists use VARs

A VAR treats every variable as potentially endogenous within a dynamic system. Each equation uses lagged values of all variables to predict one current variable.

Granger causality

Definition

In a VAR, x Granger-causes y if lagged values of x help predict y after controlling for lagged values of y and the other included variables.

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Terminology warning

Granger causality is predictive content, not structural causality.

Impulse-response intuition

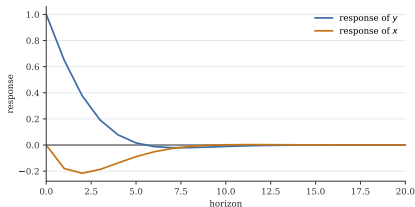
VAR(1) propagation

$$Z_t - \mu = A_1(Z_{t-1} - \mu) + e_t.$$

A one-time innovation e_t changes:

$$Z_t : e_t, \quad Z_{t+1} : A_1 e_t, \quad Z_{t+h} : A_1^h e_t.$$

IRF Recursion



Reduced-form versus structural shocks

Reduced-form innovations

The VAR residuals e_t are innovations relative to the variables and lags in the VAR. Their components may be contemporaneously correlated.

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Structural interpretation requires identification

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Practical use

VARs are useful first descriptions of multivariate dynamics and often useful forecasting tools. Policy interpretation of impulse responses requires an additional identification argument.

Five organizing applications

Application	Variables	Useful model	Main caution
Inflation persistence	inflation	AR(p), unit-root diagnostics	monetary-regime breaks
Business cycles	GDP growth, unemployment	AR, ADL, VAR	trends and revisions
Monetary policy	rates, inflation, output	VAR, impulse responses	structural identification
Exchange rates	log exchange rate, returns	random-walk benchmark	weak predictability
Asset or commodity prices	prices, returns	differenced models; volatility extensions	mean dynamics differ from volatility dynamics

How the toolkit maps to economic questions

Persistence questions

- AR models
- ACF diagnostics
- half-lives and shock decay
- regime stability checks

Trend questions

- deterministic trends
- random walks and unit roots
- transformations
- cointegration when levels move together

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- regime stability checks

Trend questions

- deterministic trends
- random walks and unit roots
- transformations
- cointegration when levels move together

Dynamic effects

Distributed lags and ADLs separate impact, delayed, and long-run effects.

Systems

VARs summarize multivariate feedback, forecasting content, and impulse-response paths.

Final takeaways

1. Time ordering makes dependence central.
2. Stationarity is the baseline language for stable dynamic behavior.
3. ARMA models describe persistence and are natural forecasting tools.
4. Unit roots require different transformations and change the meaning of shocks.
5. Dynamic regressions must be interpreted with timing and exogeneity in mind.
6. VARs summarize multivariate dynamics, but structural causal interpretation requires additional assumptions.

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6. VARs summarize multivariate dynamics, but structural causal interpretation requires additional assumptions.

Unifying perspective

Good time-series econometrics starts by asking what kind of persistence the data contain. Once we know whether shocks are temporary or permanent, whether variables are stationary or trending, and whether lags have predictive content, we can choose models that answer the economic question rather than merely fit the path.

Appendix: AR(1) mean and infinite MA representation

For

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad \mathbb{E}(\varepsilon_t) = 0,$$

stationarity implies $\mathbb{E}(Y_t) = \mathbb{E}(Y_{t-1}) = \mu$. Therefore

$$\mu = c + \phi\mu, \quad \mu = \frac{c}{1 - \phi}.$$

Centering with $y_t = Y_t - \mu$ gives

$$y_t = \phi y_{t-1} + \varepsilon_t.$$

Repeated substitution gives

$$y_t = \sum_{j=0}^m \phi^j \varepsilon_{t-j} + \phi^{m+1} y_{t-m-1}.$$

If $|\phi| < 1$ and the remote past does not dominate,

$$y_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}.$$

Appendix: AR(1) variance and ACF

Using the infinite representation and serially uncorrelated innovations,

$$\text{Var}(y_t) = \text{Var} \left(\sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j} \right) = \sum_{j=0}^{\infty} \phi^{2j} \sigma^2 = \frac{\sigma^2}{1 - \phi^2},$$

where the geometric series converges only when $|\phi| < 1$.

For $h \geq 1$,

$$\gamma_h = \text{Cov}(y_t, y_{t-h}).$$

From $y_t = \phi y_{t-1} + \varepsilon_t$,

$$\begin{aligned} \gamma_h &= \text{Cov}(\phi y_{t-1} + \varepsilon_t, y_{t-h}) \\ &= \phi \text{Cov}(y_{t-1}, y_{t-h}) + \text{Cov}(\varepsilon_t, y_{t-h}) \\ &= \phi \gamma_{h-1}. \end{aligned}$$

Thus

$$\gamma_h = \phi^h \gamma_0, \quad \rho_h = \phi^h.$$

Appendix: $AR(p)$ roots and stationarity

The $AR(p)$ model is

$$\Phi(L)Y_t = c + \varepsilon_t, \quad \Phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p.$$

For the centered process $y_t = Y_t - \mu$,

$$\Phi(L)y_t = \varepsilon_t.$$

Power-series interpretation

Stationarity requires that $\Phi(L)$ be invertible as a stable power series:

$$y_t = \Phi(L)^{-1} \varepsilon_t = \psi(L) \varepsilon_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j},$$

with absolutely summable coefficients $\sum_{j=0}^{\infty} |\psi_j| < \infty$.

Root condition

All roots z of

$$\Phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p = 0$$

must satisfy $|z| > 1$. For $AR(1)$, $z = 1/\phi$, so $|z| > 1$ is exactly $|\phi| < 1$.

Appendix: ADF regression details

Suppose the true process has higher-order autoregressive dynamics:

$$Y_t = c + \rho_1 Y_{t-1} + \cdots + \rho_p Y_{t-p} + \varepsilon_t.$$

This can be reparameterized as

$$\Delta Y_t = \alpha + \pi Y_{t-1} + \sum_{j=1}^{p-1} a_j \Delta Y_{t-j} + \varepsilon_t,$$

where

$$\pi = \rho_1 + \cdots + \rho_p - 1.$$

Test

The unit-root null is $H_0 : \pi = 0$. The lagged differences account for short-run dynamics that would otherwise appear as residual serial correlation.

Deterministic terms

No constant, a constant, or a constant plus trend imply different null and alternative models. The choice should be guided by graphs and economics, not mechanically added after seeing the result.

Appendix: ADL long-run multiplier

Start from the ADL(1,1):

$$Y_t = \alpha + \rho Y_{t-1} + \beta_0 X_t + \beta_1 X_{t-1} + u_t.$$

For a permanent change in X , set

$$Y_t = Y_{t-1} = Y, \quad X_t = X_{t-1} = X, \quad \mathbb{E}(u_t) = 0.$$

Then

$$Y = \alpha + \rho Y + (\beta_0 + \beta_1)X.$$

Move the autoregressive term to the left:

$$(1 - \rho)Y = \alpha + (\beta_0 + \beta_1)X.$$

If $|\rho| < 1$,

$$Y = \frac{\alpha}{1 - \rho} + \frac{\beta_0 + \beta_1}{1 - \rho}X.$$

Therefore the long-run multiplier is

$$\frac{\partial Y}{\partial X} = \frac{\beta_0 + \beta_1}{1 - \rho}.$$

Appendix: VAR impulse-response recursion

The VAR(p)

$$Z_t = a + A_1 Z_{t-1} + \cdots + A_p Z_{t-p} + e_t$$

can be written as a VAR(1) in companion form. Define

$$S_t = (Z'_t, Z'_{t-1}, \dots, Z'_{t-p+1})'.$$

Then

$$S_t = b + F S_{t-1} + \eta_t.$$

Impulse propagation

Ignoring later shocks, a one-time innovation at date t propagates as

$$S_{t+h} - \mathbb{E}(S_{t+h}) = F^h \eta_t.$$

For a VAR(1), this reduces to

$$Z_{t+h} - \mathbb{E}(Z_{t+h}) = A_1^h e_t.$$

Identification

Structural impulse responses also require a mapping from reduced-form innovations e_t to economically interpretable shocks.